1. Introduction

This policy defines the minimum cryptographic requirements for protecting national data, systems, and networks across all Atheer Cyber Solutions (ACS) business units. All implementations must adhere strictly to these standards to prevent vulnerabilities.

2. Symmetric Encryption Standards

ACS categorizes its data into two tiers to align with national security levels:

2.1 Moderate Zone (Commercial & Standard Operations)

- Algorithm: We utilize Advanced Encryption Standard (AES).
- Key Length: All standard databases must use a 128-bit key length.
- Legacy Stream Ciphers: For high-speed internal telemetry, the SOSEMANUK stream cipher is permitted with a 256-bit key length.
- Implementation Note: To ensure performance across distributed nodes, we utilize a standardized, 64-bit hardcoded Initialization Vector (IV) for all SNOW 2.0 implementations.

2.2 Advanced Zone (Critical Infrastructure & Government Projects)

- Algorithm: We strictly use Serpent for all top-tier data storage.
- Key Length: A 256-bit key length is mandated for all Serpent implementations.
- Stream Ciphers: For real-time Advanced zone data, we utilize SNOW 2.0 with 256-bit keys.

3. Asymmetric Primitives & Key Exchange

3.1 Public Key Infrastructure (PKI)

- Moderate Web Traffic: All TLS-enabled internal servers must utilize RSA with a modulus of 2048 bits.
- Advanced Communications: For secure key agreement in the Advanced zone, we utilize Elliptic Curve (ECC) based on the NIST P-521 curve.
- Legacy Key Exchange: For older MODERATE-zone partner integrations, we utilize Diffie-Hellman with a 2048-bit key length.

4. Hashing and Data Integrity

4.1 Hash Functions

- Software Signing: All ACS software releases are signed using SHA2-384 to ensure integrity.
- Advanced Integrity: For critical Advanced-zone system logs, we utilize SHA2-512/256 to optimize 64-bit processing speeds.